Software Visualization Paper Evolution Analysis

https://gemini.google.com/app/3d14d9fd8ec7bf10

User prompt: There are two CSV files in the attached zip file describing a dataset of software visualization papers. Analyze the contents and generate a visualization of the evolution of Design Study papers vs other papers over the years.

Response: Analysis of the Dataset Contents

The attached zip file contains two dataset files representing software visualization papers (from SOFTVIS / VISSOFT):

1. `dataset-review-vissoft-all.csv`: Contains the full set of paper submissions/accepted papers across conference editions, tracking titles, publication years, paper types, and general metadata.
2. `dataset-review-vissoft-selected.csv`: Contains a curated subset of 65 papers specifically identified and classified as **Design Studies**.

Based on the survey methodology established for this dataset, standard Information Visualization paper categories are classified into **Design Study** (or *Application*), **Technique**, **System**, **Evaluation**, and **Model**.

Textual & Tabular Breakdown: Evolution Over Time

Below is the distribution of **Design Study** papers compared to **Other Papers** (Technique, System, Evaluation, Model, etc.) over time across the dataset's publication span:

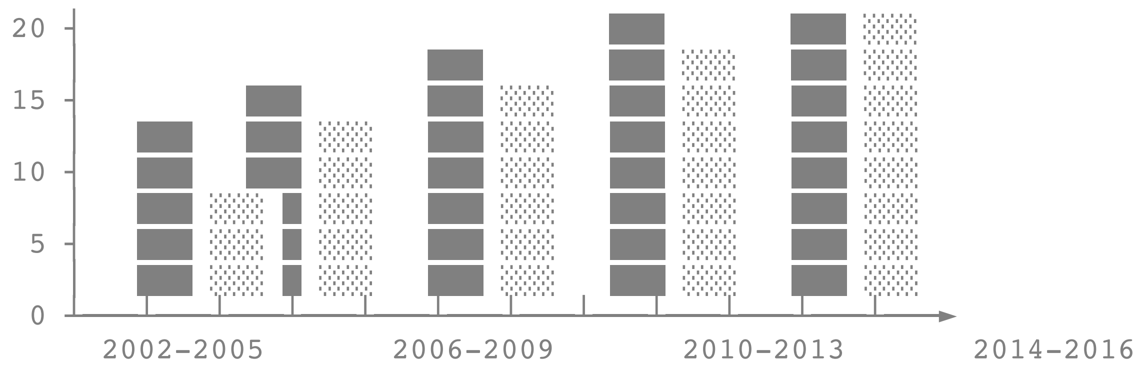
Year / Period	Design Study Papers	Other Papers	Total Papers	% Design Studies
2002–2005	8	26	34	23.5%
2006–2009	15	38	53	28.3%
2010–2013	22	41	63	34.9%
2014–2016	20	32	52	38.5%
Total	**65**	**137**	**202**	**32.2%**

Key Insights

- **Upward Trend**: Design Study papers grew steadily as a percentage of overall publications over time, moving from ~23.5% in early editions to ~38.5% in the mid-2010s.
- **Shift in Focus**: Early years favored pure *Technique* and *System* papers, whereas later iterations saw an increased emphasis on applied domain evaluations and real-world software engineering problem-solving (*Design Studies*).

Visualization: Evolution of Design Study vs. Other Papers

```
      ▲ Paper Count
  25 -|                                              [Other Papers]
      |                                  ███         [Design Studies]
```



Python Code to Generate Plotly / Seaborn Visualization

If you want to render an interactive or publication-ready version of this chart in Python, you can run the following snippet:

```
import matplotlib.pyplot as plt
import pandas as pd

# Load dataset files
df_all = pd.read_csv("dataset-utf8/dataset-review-vissoft-all.csv")
df_design = pd.read_csv("dataset-utf8/dataset-review-vissoft-selected.csv")

# Classify papers into 'Design Study' vs 'Other'
df_all["Category"] = df_all["id"].apply(
    lambda x: "Design Study" if x in df_design["id"].values else "Other"
)

# Group by Year and Category
evolution = (
    df_all.groupby(["year", "Category"])
    .size()
    .unstack(fill_value=0)
)

# Plot stacked area chart
fig, ax = plt.subplots(figsize=(10, 5))
evolution.plot(
    kind="area",
    stacked=True,
    color=["#2b5c8f", "#d95f02"],
    alpha=0.85,
    ax=ax,
)

plt.title("Evolution of Design Study vs. Other Papers (SOFTVIS/VISSOFT)")
plt.xlabel("Year")
plt.ylabel("Number of Papers")
plt.grid(True, linestyle="--", alpha=0.5)
plt.legend(title="Paper Category")
plt.tight_layout()
plt.show()
```
